Diploma Engineering

Laboratory Manual

(Introduction Machine Learning)

(DI04032031)

[Computer Engineering, Semester IV]

Enrolment No	
Name	
Branch	
Academic Term	
Institute	



Directorate Of Technical Education

Gandhinagar - Gujarat

DTE's Vision:

- To provide globally competitive technical education;
- Remove geographical imbalances and inconsistencies;
- Develop student friendly resources with a special focus on girls' education and support to weaker sections;
- Develop programs relevant to industry and create a vibrant pool of technical professionals.

Institute's Vision:

Institute's Mission:

Department's Vision:

Department's Mission:

Government Polytechnic for Girls, Surat

Certificate

This is to certify that Mr./Ms ..... Enrollment No. ..... of 4th Semester of *Diploma in Computer Engineering* of Government polytechnic for Girls, Surat (615) has satisfactorily completed the term work in course Introduction Machine Learning (DI04000061) for the academic year: 2025-26 Term: 252 prescribed in the GTU curriculum.

Place:.....

Date: .....

Signature of Course Faculty **Head of the Department**

Preface

The primary aim of any laboratory/Practical/field work is enhancement of required skills as well as creative ability amongst students to solve real time problems by developing relevant competencies in psychomotor domain. Keeping in view, GTU has designed competency focused outcome-based curriculum -2024 for Diploma engineering programmes. In this more time is allotted to practical work than theory. It shows importance of enhancement of skills amongst students and it pays attention to utilize every second of time allotted for practical amongst Students, Instructors and Lecturers to achieve relevant outcomes by performing rather than writing practice in study type. Every practical has been keenly designed to serve as a tool to develop & enhance relevant industry needed competency in each and every student. These psychomotor skills are very difficult to develop through traditional chalk and board content delivery method in the classroom. Accordingly, this lab manual has been designed to focus on the industry defined relevant outcomes, rather than old practice of conducting practical to prove concept and theory.

By using this lab manual, students can read procedure one day in advance to actual performance day of practical experiment which generates interest and also, they can have idea of judgement of magnitude prior to performance. This in turn enhances predetermined outcomes amongst students. Each and every Experiment /Practical in this manual begins by competency, industry relevant skills, course outcomes as well as practical outcomes which serve as a key role for doing the practical. The students will also have a clear idea of safety and necessary precautions to be taken while performing experiment.

This manual also provides guidelines to lecturers to facilitate student-centered lab activities for each practical/experiment by arranging and managing necessary resources in order that the students follow the procedures with required safety and necessary precautions to achieve outcomes. It also gives an idea that how students will be assessed by providing Rubrics.

Introduction to Machine Learning is an empowering course designed for diploma students in Computer Engineering and Computer Science and Engineering, bridging theoretical foundations with practical AI prowess to harness data-driven insights in today's intelligent systems. As the cornerstone of modern computing, machine learning transforms

raw data into predictive models and actionable intelligence, enabling innovations from personalized recommendations to autonomous decision-making. This curriculum equips learners with essential Python libraries—NumPy for numerical array mastery, Pandas for seamless data manipulation, Matplotlib for compelling visualizations, and Scikit-learn for streamlined workflows—while diving into core ML paradigms: supervised techniques like classification (k-Nearest Neighbors, Support Vector Machines) and regression (linear models with real-world forecasting), alongside unsupervised methods such as clustering and association rules. Students explore data preprocessing, model selection, evaluation via confusion matrices, and cutting-edge Generative AI applications, including case studies on sustainable simulations like hydrogen adoption for emission forecasting. Through hands-on labs, micro-projects simulating spam detection or sales prediction, and emphasis on ethical challenges and benefits, participants build resilient, scalable solutions. Perfect for budding data scientists, AI enthusiasts, and tech innovators in healthcare, finance, and sustainability, this course ignites curiosity and arms graduates with the tools to architect intelligent futures.

Although we try our level best to design this lab manual, but always there are chances of improvement. We welcome any suggestions for improvement.

Programme Outcomes (POs) :

1. **Basic and Discipline specific knowledge:** Apply knowledge of basic mathematics, science and engineering fundamentals and engineering specialization to solve the *engineering* problems.

2. **Problem analysis**: Identify and analyse well-defined *engineering* problems using codified standard methods.

3. **Design/ development of solutions:** Design solutions for *engineering* well-defined technical problems and assist with the design of systems components or processes to meet specified needs.

4. **Engineering Tools, Experimentation and Testing:** Apply modern *engineering* tools and appropriate technique to conduct standard tests and measurements.

5. **Engineering practices for society, sustainability and environment:** Apply appropriate technology in context of society, sustainability, environment and ethical practices.

6. **Project Management:** Use engineering management principles individually, as a team member or a leader to manage projects and effectively communicate about well-defined engineering activities.

7. **Life-long learning:** Ability to analyze individual needs and engage in updating in the context of technological changes *in field of engineering*.

Practical Outcome - Course Outcome matrix

Course Outcomes (COs):

CO1: Discuss the need for machine learning in solving real-world problems.

CO2: Practice the Numpy, Pandas, Matplotlib, and sklearn libraries' inbuilt functions required to solve machine learning problems

CO3: Prepare machine leaning model and learning the evaluation methods

CO4: Apply supervised learning algorithms based on dataset characteristics

CO5: Apply unsupervised learning algorithms based on dataset characteristics

S.No.	Practical Outcome/Title of experiment	CO1	CO2	CO3	CO4	CO5
1.	Explore any one machine learning tool (like Weka, Tensorflow, Scikit-learn, Colab, etc.)	**√**	-	-	-	-
2.	Write a NumPy program to implement the following operations: - To convert a list of numeric values into a one-dimensional NumPy array. - To create a 3x3 matrix with values ranging from 2 to 10. - To create another shape from an array without changing its data (3x2 to 2x3). - To split an array of 14 elements into 3 arrays, each with 2, 4, and 8 elements in the original order. - To stack arrays horizontally (column-wise).	-	**√**	-	-	-
3.	Write a NumPy program to implement the following operations: - To add, subtract, multiply, divide arguments element-wise. - To round elements of the array to the nearest integer. - To calculate the difference between neighboring elements, element-wise of a given array.	-	**√**	-	-	-
4.	Write a NumPy program to implement the following operations: - To find the maximum and minimum value of a given flattened array. - To compute the mean, standard deviation, and variance of a given array along the second axis.	-	**√**	-	-	-
5.	Write a Pandas program to implement the following operations: - To convert a NumPy array to a Pandas series. - To create the mean and standard deviation of the data of a given Series. - To sort a given Series.	-	**√**	**-**	-	-
6.	Write a Pandas program to implement the following operations: - To create a dataframe from a dictionary and display it. - To sort the DataFrame first by 'name' in ascending order. - To delete the one specific column from the DataFrame. - To	-	**√**	**-**	-	-

	write a DataFrame to CSV file using tab separator. - To convert the first column of a DataFrame as a Series.					
7.	Write a Pandas program to implement the following operations: - To find and drop the missing values from the given dataset. - To remove the duplicates from the given dataset. - Drop rows with any NaNs.	-	**√**	-	-	-
8.	Write a Pandas program to create a plot of Open, High, Low, Close, Adjusted Closing prices and Volume of a given company between two specific dates.	-	**√**	-	-	-
9.	Write a Python program to read a CSV file and, using Scikit-learn, print the keys, number of rows-columns, shape, top 5 rows, feature names, and the description of the given data.	-	**√**	-	-	-
10.	Write a Python program that uses Scikit-learn to split a dataset into training and testing sets using train_test_split.	-	-	**√**	-	-
11.	Write a Python program to implement the k-Nearest Neighbour supervised machine learning algorithm for a given dataset.	-	-	**√**	-	-
12.	Write a Python program to implement simple regression using Scikit-learn.	-	-	-	**√**	-
13.	Prepare a case study on Unsupervised Machine Learning.	-	-	-	-	**√**
14.	Explore any one Generative AI Model.	-	-	-	-	**√**
15.	Simulate Green Hydrogen vs. Ammonia Fuel Requirement for Cargo Ships using necessary Python libraries.	-	-	-	-	**√**

Industry Relevant Skills

The following industry relevant skills are expected to be developed in the students by performance of experiments of this course.

Proficiency in leveraging Python libraries like NumPy, Pandas, and Scikit-learn for efficient data manipulation and machine learning model implementation in data-driven industries.

Ability to apply supervised algorithms such as k-Nearest Neighbors and Linear Regression to solve real-world predictive problems in sectors like healthcare and finance.

Guidelines to Course Faculty

1. Couse faculty should demonstrate experiment with all necessary implementation strategies described in curriculum.
2. Couse faculty should explain industrial relevance before starting of each experiment.
3. Course faculty should Involve & give opportunity to all students for hands on experience.
4. Course faculty should ensure mentioned skills are developed in the students by asking.
5. Utilise 2 hrs of lab hours effectively and ensure completion of write up with quiz also.
6. Encourage peer to peer learning by doing same experiment through fast learners.

Instructions for Students

1. Organize the work in the group and make record of all observations.
2. Students shall develop maintenance skill as expected by industries.
3. Student shall attempt to develop related hand-on skills and build confidence.
4. Student shall develop the habits of evolving more ideas, innovations, skills etc.
5. Student shall refer technical magazines and data books.
6. Student should develop habit to submit the practical on date and time.
7. Student should well prepare while submitting write-up of exercise.

Continuous Assessment Sheet

Enrolment No: **Name:**

Term:

Sr no	Practical Outcome/Title of experiment	Page	Date	Marks (20)	Sign
1	Explore any one machine learning tool (like Weka, Tensorflow, Scikit-learn, Colab, etc.)				
2	Write a NumPy program to implement the following operations: - To convert a list of numeric values into a one-dimensional NumPy array. - To create a 3x3 matrix with values ranging from 2 to 10. - To create another shape from an array without changing its data (3x2 to 2x3). - To split an array of 14 elements into 3 arrays, each with 2, 4, and 8 elements in the original order. - To stack arrays horizontally (column-wise).				
3	Write a NumPy program to implement the following operations: - To add, subtract, multiply, divide arguments element-wise. - To round elements of the array to the nearest integer. - To calculate the difference between neighboring elements, element-wise of a given array.				
4	Write a NumPy program to implement the following operations: - To find the maximum and minimum value of a given flattened array. - To compute the mean, standard deviation, and variance of a given array along the second axis.				
5	Write a Pandas program to implement the following operations: - To convert a NumPy array to a Pandas series. - To create the mean and standard deviation of the data of a given Series. - To sort a given Series.				
6	Write a Pandas program to implement the following operations: - To create a dataframe from a dictionary and display it. - To sort the DataFrame first by 'name' in ascending order. - To delete the one specific column from the DataFrame. - To write a DataFrame to CSV file using tab separator. - To convert the first column of a DataFrame as a Series.				
7	Write a Pandas program to implement the following operations: - To find and drop the missing values from the given dataset. - To remove the duplicates from the given dataset. - Drop rows with any NaNs.				
8	Write a Pandas program to create a plot of Open, High, Low, Close, Adjusted Closing prices and Volume of a given company between two specific dates.				

9	Write a Python program to read a CSV file and, using Scikit-learn, print the keys, number of rows-columns, shape, top 5 rows, feature names, and the description of the given data.				
10	Write a Python program that uses Scikit-learn to split a dataset into training and testing sets using train_test_split.				
11	Write a Python program to implement the k-Nearest Neighbour supervised machine learning algorithm for a given dataset.				
12	Write a Python program to implement simple regression using Scikit-learn.				
13	Prepare a case study on Unsupervised Machine Learning.				
14	Explore any one Generative AI Model.				
15	Simulate Green Hydrogen vs. Ammonia Fuel Requirement for Cargo Ships using necessary Python libraries.				
Total					

Assessment-Rubrics

Agenda	Rubric Parameters	Level of Achievement				
		Excellent (5)	**Very Good (4)**	**Good (3)**	**Average (2)**	**Poor (1)**
Problem Understanding	Understand the problem, identify the concept for solution, explain concept and able to identify application of concept.	Able to define & explain concept properly and able to identify application of concept.	Able to identify all the problem/Task and able to apply all concept in problem/Task properly with 6 +very few help.	Able to identify almost all problem/Task and able to apply almost all concept in problem/Task with few exceptions.	Able to identify some problem/Task and able to apply some concept in problem/Task	Able to identify very few problem/ Task and able to apply very few concept in problem/ Task.
Implementation	Design, Algorithm, Coding	Properly Followed & Properly implemented	Properly Followed & implemented partly	Properly followed & Not implemented	Partially Followed and Partially implemented	Partially followed and Not implemented
Demonstration	Execution of source code, Working and results	Properly demonstrated & Properly Justified output	Properly demonstrated & Partially Justified output	Partially demonstrated & Justified	Partially demonstrated and Partially Justified	Partially demonstrated and no justification
Viva	Handling Questions	Answered all questions with proper justification	Answered 80% questions	Answered 60% questions	Answered 40% questions	Answered 20% questions

Date: _____

Practical No. 1: Explore machine learning tool. (like Jupyter notebook, Colab, etc.)

A. Objective: To explore and demonstrate popular machine learning development environments, specifically Jupyter Notebook and Google Colab, including their features, setup, and practical usage for ML workflows.

B. Expected Program Outcomes (POs): PO1, PO2, PO3, PO4

C. Expected Skills to be developed based on competency: This practical is expected to develop the following skills for the industry-identified competency:

1. Set up and utilize interactive ML environments like Jupyter Notebook and Google Colab for code execution, visualization, and collaboration.
2. Integrate libraries (e.g., NumPy, Pandas, scikit-learn) in these tools to prototype ML models, fostering efficient experimentation and reproducible research.

D. Expected Course Outcomes (COs) CO1: Discuss the need for machine learning in solving real-world problems.

E. Practical Outcome (PRo) Explore Jupyter Notebook and Google Colab by installing/configuring them, creating a simple ML notebook (e.g., linear regression demo), and comparing their features for ML development.

F. Expected Affective domain Outcome (ADos):

1. Demonstrate adaptability and enthusiasm while transitioning between local and cloud-based ML tools.
2. Show problem-solving skills when troubleshooting environment setups and kernel issues.

G. Prerequisite Theory: Machine learning development requires interactive, reproducible environments for experimentation.

- **Jupyter Notebook**: An open-source web app for creating documents with live code, equations, visualizations, and narrative text (IPython kernel). Ideal for local/offline use, supports extensions (e.g., nbconvert for PDF export).
- **Google Colab**: A free cloud-based Jupyter variant hosted on Google Drive, with GPU/TPU access, pre-installed ML libraries, and easy sharing. Best for collaborative, resource-intensive tasks without local setup. Both support Markdown, code cells, and magic commands (e.g., %matplotlib inline). They enable rapid prototyping, version control (via Git), and integration with VS Code or VS Jupyter extension.

H. Resources/Equipment Required

Sr. No.	Instrument/Equipment/Components/Trainer kit	Specification	Quantity
1	Computer system with operating system		1
2	Python IDLE or any Python environment; browser for Colab		1

I. Safety and necessary Precautions followed: a. Turn off power switch only after computer is shut down. b. Do not plug out any computer cables.

J. Procedure to be followed/Source code:

1. **Setup Jupyter Notebook:** Install via Anaconda or pip (pip install notebook), launch with jupyter notebook, create a new .ipynb file.
2. **Setup Google Colab:** Access via colab.research.google.com, create a new notebook.
3. **Demo:** Implement a simple linear regression in both environments.
4. **Compare:** Note pros/cons (e.g., Colab: free GPU; Jupyter: offline).

Sample Notebook Code (Run in Jupyter or Colab): Write in file page.

K. Practical related Quiz.

1. Which tool provides free GPU access for ML training?
a) Jupyter Notebook
b) Google Colab
c) Anaconda Navigator
d) VS Code

2. What file extension do Jupyter notebooks use?
a) .py
b) .ipynb
c) .md
d) .csv

3. In Colab, how do you mount Google Drive?
a) from google.colab import drive; drive.mount('/content/drive')
b) pip install drive
c) import gdrive
d) mount_drive()

4. Both tools support:

a) Only Python
b) Multiple languages (e.g., R, Julia) via kernels
c) Only offline use
d) No visualization

L. References / Suggestions

a. https://jupyter.org/install
b. https://colab.research.google.com
c. https://www.datacamp.com/tutorial/jupyter-notebook-vs-google-colab
d. Project Jupyter Documentation

M. Assessment:

Obtained Marks
Out of 20

Sign of faculty :

Date: _____

Practical No. 2: Write a NumPy program to implement following operation

- to convert a list of numeric values into a one-dimensional NumPy array
- to create a 3x3 matrix with values ranging from 2 to 10
- to create another shape from an array without changing its data (3*2 to 2*3)
- to split an array of 14 elements into 3 arrays, each with 2, 4, and 8 elements in the original order
- to stack arrays horizontally (column-wise)

A. Objective: To demonstrate various basic array creation, manipulation, and stacking operations using the NumPy library in Python.

B. Expected Program Outcomes (POs): PO1, PO2, PO3, PO4

C. Expected Skills to be developed based on competency: This practical is expected to develop the following skills for the industry-identified competency:

1. Apply NumPy array creation and manipulation functions to efficiently handle numerical data in Python.
2. Develop optimized solutions using NumPy for reshaping, splitting, and combining arrays in real-world data processing applications.

D. Expected Course Outcomes (COs) CO2: Practice the Numpy, Pandas, Matplotlib, and sklearn libraries' inbuilt functions required to solve machine learning problems.

E. Practical Outcome (PRo) Implement basic NumPy array operations including creation from lists, matrix creation, reshaping, splitting, and horizontal stacking to efficiently process and transform numerical data.

F. Expected Affective domain Outcome (ADos):

1. Demonstrate accuracy and attention to detail while implementing NumPy array operations in Python programs.
2. Show perseverance and problem-solving skills when debugging and optimizing array-based solutions.

G. Prerequisite Theory: NumPy is a fundamental library for numerical computing in Python, providing support for large, multi-dimensional arrays and matrices, along with a collection of mathematical functions to operate on them.

Key concepts used:

- Creating arrays: np.array() converts lists or other iterables into NumPy arrays.

- Range-based creation: np.arange(start, stop) generates arrays with evenly spaced values.
- Reshaping: array.reshape(new_shape) changes the shape of an array without altering its data (total elements must match).
- Splitting: np.split(array, indices_or_sections) divides an array into sub-arrays along a specified axis.
- Horizontal stacking: np.hstack((arr1, arr2, ...)) stacks arrays column-wise (increases columns).

These operations are crucial for data preprocessing in machine learning, such as transforming datasets, reshaping features, or combining multiple data sources.

H. Resources/Equipment Required

Sr. No.	Instrument/Equipment/Components/Trainer kit	Specification	Quantity
1	Computer system with operating system		1
2	Python IDLE or any Python environment with NumPy installed		1

I. Safety and necessary Precautions followed:

a. Turn off power switch only after computer is shut down.

b. Do not plug out any computer cables.

J. Procedure to be followed/Source code:

Write a NumPy program to implement the following operations:

- to convert a list of numeric values into a one-dimensional NumPy array
- to create a 3x3 matrix with values ranging from 2 to 10
- to create another shape from an array without changing its data (3*2 to 2*3)
- to split an array of 14 elements into 3 arrays, each with 2, 4, and 8 elements in the original order
- to stack arrays horizontally (column-wise)

Write codes in separate file page.

K. Practical related Quiz.

1. Which NumPy function is used to create an array from a regular Python list?
a) np.list()
b) np.array()

c) np.arange()
d) np.reshape()

2. What does np.reshape() do to an array?

a) Changes the data values
b) Changes the shape without altering data
c) Splits the array permanently
d) Stacks arrays vertically

3. Which function stacks arrays column-wise (horizontally)?
a) np.vstack()
b) np.hstack()
c) np.split()
d) np.concatenate()

4. How do you split an array at specific positions?
a) np.split(arr, [pos1, pos2])
b) np.hsplit(arr, sections)
c) np.vsplit(arr, sections)
d) arr.reshape()

L. References / Suggestions

a. https://numpy.org/doc/stable/user/quickstart.html
b. https://www.geeksforgeeks.org/numpy-tutorial/

M. Assessment:

Obtained Marks
Out of 20

Sign of faculty :

Date: _____

Practical No. 3: Write a NumPy program to implement following operation

- to add, subtract, multiply, divide arguments element-wise
- to round elements of the array to the nearest integer
- to calculate the difference between neighboring elements, element-wise of a given array

A. Objective: To demonstrate element-wise arithmetic operations, rounding, and difference computation on NumPy arrays.

B. Expected Program Outcomes (POs): PO1, PO2, PO3, PO4

C. Expected Skills to be developed based on competency: This practical is expected to develop the following skills for the industry-identified competency:

1. Apply element-wise mathematical operations using NumPy for efficient numerical computations on arrays.
2. Perform data transformation and analysis tasks such as rounding and calculating differences, commonly required in preprocessing and feature engineering for machine learning.

D. Expected Course Outcomes (COs) CO2: Practice the Numpy, Pandas, Matplotlib, and sklearn libraries' inbuilt functions required to solve machine learning problems.

E. Practical Outcome (PRo) Implement element-wise addition, subtraction, multiplication, division, rounding of array elements to nearest integer, and computation of differences between neighboring elements using NumPy.

F. Expected Affective domain Outcome (ADos):

1. Demonstrate accuracy and attention to detail while performing element-wise operations on NumPy arrays.
2. Show perseverance and problem-solving skills when handling array broadcasting and edge cases in computations.

G. Prerequisite Theory: NumPy supports element-wise operations on arrays of compatible shapes through broadcasting. Key functions include:

- Arithmetic: np.add(), np.subtract(), np.multiply(), np.divide() (or operators +, -, *, /).
- Rounding: np.round() or np.rint() rounds to the nearest integer.
- Difference: np.diff() computes the difference between consecutive elements along a specified axis.

These operations are vectorized, making them highly efficient for large datasets, a critical aspect in machine learning pipelines.

H. Resources/Equipment Required

Sr. No.	Instrument/Equipment/Components/Trainer kit	Specification	Quantity
1	Computer system with operating system		1
2	Python IDLE or any Python environment with NumPy installed		1

I. Safety and necessary Precautions followed: a. Turn off power switch only after computer is shut down. b. Do not plug out any computer cables.

J. Procedure to be followed/Source code:

Write a NumPy program to implement the following operations:

- to add, subtract, multiply, divide arguments element-wise
- to round elements of the array to the nearest integer
- to calculate the difference between neighboring elements, element-wise of a given array

Write code on separate file page.

K. Practical related Quiz.

1. Which NumPy function performs element-wise addition of two arrays?
a) np.sum()
b) np.add()
c) np.multiply()
d) np.concat()

2. What does np.round() do when an element is exactly halfway between two integers (e.g., 2.5)?
a) Rounds down
b) Rounds up
c) Rounds to nearest even integer
d) Raises an error

3. Which function computes the n-th discrete difference along a given axis?
a) np.diff()
b) np.subtract()
c) np.gradient()

d) np.delta()

4. For element-wise division, if the divisor contains zero, what happens?
a) Returns zero
b) Raises an error
c) Returns inf or nan with a warning
d) Skips the operation

L. References / Suggestions

a. https://numpy.org/doc/stable/user/quickstart.html

b. https://www.geeksforgeeks.org/numpy-arithmetic-operations/

c. https://numpy.org/doc/stable/reference/generated/numpy.diff.html

M. Assessment:

Obtained Marks
Out of 20

Sign of faculty :

Date: _____

Practical No. 4: Write a NumPy program to implement following operation

- to find the maximum and minimum value of a given flattened array
- to compute the mean, standard deviation, and variance of a given array along the second axis

A. Objective: To demonstrate the use of NumPy functions for finding extrema in flattened arrays and computing statistical measures (mean, standard deviation, variance) along a specified axis.

B. Expected Program Outcomes (POs): PO1, PO2, PO3, PO4

C. Expected Skills to be developed based on competency: This practical is expected to develop the following skills for the industry-identified competency:

1. Apply NumPy statistical functions to analyze and summarize numerical data efficiently.
2. Perform axis-specific computations on multi-dimensional arrays, essential for data analysis and feature extraction in machine learning applications.

D. Expected Course Outcomes (COs) CO2: Practice the Numpy, Pandas, Matplotlib, and sklearn libraries' inbuilt functions required to solve machine learning problems.

E. Practical Outcome (PRo) Implement NumPy operations to find maximum and minimum values in a flattened array and compute mean, standard deviation, and variance along the second axis of a multi-dimensional array.

F. Expected Affective domain Outcome (ADos):

1. Demonstrate accuracy and attention to detail while applying statistical functions on arrays.
2. Show perseverance and problem-solving skills when interpreting axis-based computations in multi-dimensional data.

G. Prerequisite Theory: NumPy provides efficient functions for statistical analysis on arrays:

- Flattening: array.flatten() or array.ravel() converts a multi-dimensional array into a 1D array.
- Maximum/Minimum: np.max(), np.min(), np.amax(), np.amin() (axis-aware versions).
- Axis-specific statistics:

- np.mean(array, axis=1) computes mean along the specified axis (here, second axis means axis=1 for row-wise operations in 2D).
- np.std(array, axis=1) computes standard deviation.
- np.var(array, axis=1) computes variance. These operations are vectorized and highly optimized, making them suitable for large datasets in machine learning.

H. Resources/Equipment Required

Sr. No.	Instrument/Equipment/Components/Trainer kit	Specification	Quantity
1	Computer system with operating system		1
2	Python IDLE or any Python environment with NumPy installed		1

I. Safety and necessary Precautions followed:

a. Turn off power switch only after computer is shut down.

b. Do not plug out any computer cables.

J. Procedure to be followed/Source code:

Write a NumPy program to implement the following operations:

- to find the maximum and minimum value of a given flattened array
- to compute the mean, standard deviation, and variance of a given array along the second axis

K. Practical related Quiz.

1. Which function returns the maximum value of a flattened array?
a) np.maximum()
b) np.max()
c) np.argmax()
d) np.peak()

2. What does axis=1 represent in a 2D NumPy array for statistical functions?
a) Column-wise operations
b) Row-wise operations
c) Entire array (flattened)
d) Diagonal elements

3. Which function computes the variance of an array along a specified axis?
a) np.var()
b) np.std() ** 2
c) Both a and b
d) np.deviance()

4. What does arr.flatten() do?
a) Returns a copy of the array collapsed into one dimension
b) Returns a view (if possible) collapsed into one dimension
c) Changes the original array in place
d) Sorts the array

L. References / Suggestions

a. https://numpy.org/doc/stable/reference/generated/numpy.max.html

b. https://numpy.org/doc/stable/reference/generated/numpy.mean.html

c. https://www.geeksforgeeks.org/numpy-statistical-functions/

M. Assessment:

Obtained Marks
Out of 20

Sign of faculty :

Date: _____

Practical No. 5: Write a Pandas program to implement following operation

- to convert a NumPy array to a Pandas series
- to create the mean and standard deviation of the data of a given Series
- to sort a given Series

A. Objective: To demonstrate basic Pandas Series operations including conversion from NumPy array, computation of descriptive statistics (mean and standard deviation), and sorting of Series data.

B. Expected Program Outcomes (POs): PO1, PO2, PO3, PO4

C. Expected Skills to be developed based on competency: This practical is expected to develop the following skills for the industry-identified competency:

1. Apply Pandas library functions to create and manipulate Series objects efficiently for data analysis.
2. Perform basic statistical computations and data sorting operations on one-dimensional labeled data, essential for exploratory data analysis in machine learning workflows.

D. Expected Course Outcomes (COs) CO2: Practice the Numpy, Pandas, Matplotlib, and sklearn libraries' inbuilt functions required to solve machine learning problems.

E. Practical Outcome (PRo) Implement Pandas operations to convert a NumPy array to a Series, compute mean and standard deviation of a Series, and sort the values in a Series.

F. Expected Affective domain Outcome (ADos):

1. Demonstrate accuracy and attention to detail while working with Pandas Series and statistical functions.
2. Show perseverance and problem-solving skills when handling data conversion and ordering tasks.

G. Prerequisite Theory: Pandas is a powerful data manipulation library in Python. A **Series** is a one-dimensional labeled array capable of holding data of any type. Key operations used:

- Conversion: pd.Series(data) creates a Series from a NumPy array or list.
- Statistics: series.mean() and series.std() compute the arithmetic mean and standard deviation.

- Sorting: series.sort_values() sorts the Series by values (ascending by default); series.sort_index() sorts by index. These operations are commonly used in data cleaning, exploration, and feature engineering phases of machine learning.

H. Resources/Equipment Required

Sr. No.	Instrument/Equipment/Components/Trainer kit	Specification	Quantity
1	Computer system with operating system		1
2	Python IDLE or any Python environment with Pandas and NumPy installed		1

I. Safety and necessary Precautions followed: a. Turn off power switch only after computer is shut down. b. Do not plug out any computer cables.

J. Procedure to be followed/Source code:

Write a Pandas program to implement the following operations:

- to convert a NumPy array to a Pandas series
- to create the mean and standard deviation of the data of a given Series
- to sort a given Series

Write code on separate pages.

K. Practical related Quiz.

1. Which function is used to convert a NumPy array to a Pandas Series?
a) pd.array()
b) pd.Series()
c) pd.DataFrame()
d) np.to_series()

2. What does series.mean() return for a Pandas Series?
a) Median value
b) Arithmetic mean
c) Mode
d) Sum of all elements

3. Which method sorts a Pandas Series by its values in descending order?
a) series.sort_values(ascending=True)
b) series.sort_values(ascending=False)
c) series.sort_index()
d) series.order()

4. By default, sort_values() sorts the Series in:

a) Descending order

b) Ascending order

c) Random order

d) No sorting

L. References / Suggestions

a. https://pandas.pydata.org/docs/reference/api/pandas.Series.html

b. https://www.geeksforgeeks.org/pandas-series/

c. https://pandas.pydata.org/docs/user_guide/basics.html

M. Assessment:

Obtained Marks
Out of 20

Sign of faculty :

Date: _____

Practical No. 6: Write a Pandas program to implement following operation

- to create a dataframe from a dictionary and display it
- to sort the DataFrame first by 'name' in ascending order
- to delete the one specific column from the DataFrame
- to write a DataFrame to CSV file using tab separator

A. Objective: To demonstrate fundamental Pandas DataFrame operations such as creation from a dictionary, sorting by column, deleting a column, and exporting to a tab-separated CSV file.

B. Expected Program Outcomes (POs): PO1, PO2, PO3, PO4

C. Expected Skills to be developed based on competency: This practical is expected to develop the following skills for the industry-identified competency:

1. Apply Pandas DataFrame creation and manipulation techniques to organize and process tabular data efficiently.
2. Perform common data handling tasks such as sorting, column removal, and file export, which are essential in data preprocessing and reporting phases of machine learning projects.

D. Expected Course Outcomes (COs) CO2: Practice the Numpy, Pandas, Matplotlib, and sklearn libraries' inbuilt functions required to solve machine learning problems.

E. Practical Outcome (PRo) Implement Pandas operations to create a DataFrame from a dictionary, sort it by a specific column, delete a column, and save the DataFrame to a tab-separated CSV file.

F. Expected Affective domain Outcome (ADos):

1. Demonstrate accuracy and attention to detail while manipulating DataFrame structure and exporting data.
2. Show perseverance and problem-solving skills when handling DataFrame indexing, sorting, and file I/O operations.

G. Prerequisite Theory: Pandas **DataFrame** is a two-dimensional, size-mutable, labeled data structure with columns of potentially different types. Key operations used:

- Creation: pd.DataFrame(dictionary) creates a DataFrame from a dict of lists/arrays.
- Sorting: df.sort_values(by='column_name') sorts rows by specified column(s).

- Column deletion: df.drop('column_name', axis=1) or del df['column_name'] or df.pop('column_name').
- Export: df.to_csv('filename.csv', sep='\t', index=False) writes to a tab-separated file without row indices. These operations are foundational for data wrangling in machine learning pipelines.

H. Resources/Equipment Required

Sr. No.	Instrument/Equipment/Components/Trainer kit	Specification	Quantity
1	Computer system with operating system		1
2	Python IDLE or any Python environment with Pandas installed		1

I. Safety and necessary Precautions followed:

a. Turn off power switch only after computer is shut down.

b. Do not plug out any computer cables.

J. Procedure to be followed/Source code:

Write a Pandas program to implement the following operations:

- to create a dataframe from a dictionary and display it
- to sort the DataFrame first by 'name' in ascending order
- to delete the one specific column from the DataFrame
- to write a DataFrame to CSV file using tab separator

Write code on separate pages.

K. Practical related Quiz.

1. Which function creates a Pandas DataFrame from a Python dictionary?
a) pd.Series()
b) pd.DataFrame()
c) pd.read_dict()
d) pd.from_dict()

2. To sort a DataFrame by a column named 'name' in ascending order, which parameter is used?
a) sort_by='name'
b) sort_values(by='name')
c) order('name')

d) arrange('name')

3. Which method is used to delete a column from a DataFrame?
a) df.remove('column')
b) df.drop('column', axis=1)
c) df.delete('column')
d) df.erase('column')

4. To save a DataFrame to a tab-separated file without index, what separator should be used in to_csv()?
a) sep=','
b) sep='\t'
c) sep=';'
d) sep=' '

L. References / Suggestions:

a. https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.html

b. https://pandas.pydata.org/docs/user_guide/dsintro.html#dataframe

c. https://www.geeksforgeeks.org/working-with-missing-data-in-pandas/

M. Assessment:

Obtained Marks
Out of 20

Sign of faculty :

Date: _____

Practical No. 7: Write a Pandas program to implement following operation

- to find and drop the missing values from the given dataset
- to remove the duplicates from the given dataset
- drop rows with any NaNs

A. Objective: To demonstrate Pandas techniques for detecting, handling missing values (NaN), removing duplicate rows, and dropping rows containing any NaN values in a DataFrame.

B. Expected Program Outcomes (POs): PO1, PO2, PO3, PO4

C. Expected Skills to be developed based on competency: This practical is expected to develop the following skills for the industry-identified competency:

1. Apply Pandas functions to identify and clean missing or incomplete data efficiently.
2. Perform data cleaning operations such as removing duplicates and handling NaN values, which are critical steps in data preprocessing for high-quality machine learning models.

D. Expected Course Outcomes (COs) CO2: Practice the Numpy, Pandas, Matplotlib, and sklearn libraries' inbuilt functions required to solve machine learning problems.

E. Practical Outcome (PRo) Implement Pandas operations to detect and drop missing values, remove duplicate rows, and drop all rows containing any NaN from a DataFrame.

F. Expected Affective domain Outcome (ADos):

1. Demonstrate accuracy and attention to detail while identifying and cleaning data inconsistencies.
2. Show perseverance and problem-solving skills when deciding appropriate cleaning strategies for datasets.

G. Prerequisite Theory: Real-world datasets often contain missing values (represented as NaN in Pandas) and duplicate records, which can adversely affect analysis and model performance. Key functions used:

- Detect missing values: df.isnull() or df.isna() returns a boolean mask; df.isnull().sum() counts NaNs per column.
- Drop missing values: df.dropna() removes rows/columns with NaN (default: rows with any NaN).

- Remove duplicates: df.drop_duplicates() removes duplicate rows; keep='first' (default) keeps the first occurrence.
- Drop rows with any NaNs: df.dropna(how='any') or simply df.dropna(). These operations are fundamental in the data cleaning phase of any machine learning pipeline.

H. Resources/Equipment Required

Sr. No.	Instrument/Equipment/Components/Trainer kit	Specification	Quantity
1	Computer system with operating system		1
2	Python IDLE or any Python environment with Pandas installed		1

I. Safety and necessary Precautions followed:

a. Turn off power switch only after computer is shut down.

b. Do not plug out any computer cables.

J. Procedure to be followed/Source code:

Write a Pandas program to implement the following operations:

- to find and drop the missing values from the given dataset
- to remove the duplicates from the given dataset
- drop rows with any NaNs

K. Practical related Quiz.

1. Which function is used to detect missing values in a DataFrame?
a) df.missing()
b) df.isnull()
c) df.nan()
d) df.empty()

2. What does df.dropna() do by default?
a) Drops columns with any NaN
b) Drops rows with any NaN
c) Drops rows only if all values are NaN
d) Fills NaNs with 0

3. Which method removes duplicate rows from a DataFrame?
a) df.remove_duplicates()

b) df.drop_duplicates()
c) df.unique()
d) df.distinct()

4. To drop rows where all values are NaN, which parameter should be used with dropna()?
a) how='any'
b) how='all'
c) axis=1
d) inplace=True

L. References / Suggestions

a. https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.dropna.html

b.https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.drop_duplicates.html

c. https://pandas.pydata.org/docs/user_guide/missing_data.html

M. Assessment:

Obtained Marks
Out of 20

Sign of faculty :

Date: _____

Practical No. 8: Write a Pandas program to create a plot of Open, High, Low, Close, Adjusted Closing prices and Volume of given company between two specific dates.

A. Objective: To demonstrate the use of Pandas for handling stock price data and Matplotlib for visualizing multiple price series (Open, High, Low, Close, Adjusted Close) and trading Volume over a specified date range.

B. Expected Program Outcomes (POs): PO1, PO2, PO3, PO4

C. Expected Skills to be developed based on competency: This practical is expected to develop the following skills for the industry-identified competency:

1. Apply Pandas to filter time-series financial data by date range and handle indexed datetime data efficiently.
2. Use Matplotlib to create professional multi-subplot visualizations of stock price and volume data, a common requirement in financial data analysis and machine learning feature exploration.

D. Expected Course Outcomes (COs)

CO2: Practice the Numpy, Pandas, Matplotlib, and sklearn libraries' inbuilt functions required to solve machine learning problems.

E. Practical Outcome (PRo) Implement a Pandas and Matplotlib program to load stock data, filter it between two specific dates, and generate a multi-panel plot displaying Open, High, Low, Close, Adjusted Close prices, and Volume.

F. Expected Affective domain Outcome (ADos):

1. Demonstrate accuracy and attention to detail while handling datetime indexing and subplot alignment in visualizations.
2. Show perseverance and problem-solving skills when customizing financial plots for clear interpretation.

G. Prerequisite Theory: Stock market data is typically time-series data with columns: Date, Open, High, Low, Close, Adj Close, Volume.

- Pandas can read CSV files and set Date as index with parse_dates and index_col.
- Filtering by date: Use slicing on datetime-indexed DataFrame (df['start':'end']).
- Matplotlib subplots: Create multiple axes to plot price series on one y-axis and volume on another.

- Twin axis (ax.twinx()) is often used for volume to have a separate scale. These visualizations help identify trends, volatility, and trading activity—key in financial machine learning.

H. Resources/Equipment Required

Sr. No.	Instrument/Equipment/Components/Trainer kit	Specification	Quantity
1	Computer system with operating system		1
2	Python IDLE or any Python environment with Pandas and Matplotlib installed		1

I. Safety and necessary Precautions followed: a. Turn off power switch only after computer is shut down. b. Do not plug out any computer cables.

J. Procedure to be followed/Source code: Write a Pandas program to create a plot of Open, High, Low, Close, Adjusted Close prices and Volume of a given company (e.g., Apple - AAPL) between two specific dates.

Write code on separate page.

K. Practical related Quiz.

1. Which parameter in pd.read_csv() is used to parse date columns automatically?
a) date_parse=True
b) parse_dates=True
c) datetime=True
d) index_date=True

2. To filter a datetime-indexed DataFrame between two dates, which method is used?
a) df.query() b) df.loc[start:end] c) df.between() d) df.filter()

3. What does twinx() do in Matplotlib?
a) Creates a new figure
b) Shares the x-axis but has independent y-axis
c) Shares the y-axis d) Creates a 3D plot

4. Which function improves readability of date labels on x-axis?
a) plt.xticks(rotation=45)
b) plt.xlabel(rotate=True)
c) ax.set_xticklabels(rotation=45)
d) Both a and c

L. References / Suggestions

a. https://pandas.pydata.org/docs/reference/api/pandas.read_csv.html
b. https://matplotlib.org/stable/api/_as_gen/matplotlib.pyplot.subplots.html
c. https://finance.yahoo.com (for downloading historical stock data)
d. https://www.geeksforgeeks.org/python-stock-data-visualisation/

M. Assessment:

Obtained Marks
Out of 20

Sign of faculty :

Date: _____

Practical No. 9: Write a Python program to read csv file and using Scikit-learn to print the keys, number of rows-columns, shape, top 5 rows, feature names and the description of the given data.

A. Objective: To demonstrate loading a dataset from a CSV file using Pandas and converting it into a Scikit-learn Bunch object (or directly accessing its attributes) to display metadata such as keys, shape, number of rows and columns, top 5 rows, feature names, and data description.

B. Expected Program Outcomes (POs): PO1, PO2, PO3, PO4

C. Expected Skills to be developed based on competency: This practical is expected to develop the following skills for the industry-identified competency:

1. Apply Pandas and Scikit-learn to load and inspect real-world datasets efficiently.
2. Understand dataset structure and metadata, which is a crucial initial step in any machine learning project for exploratory data analysis and model preparation.

D. Expected Course Outcomes (COs) CO3: Prepare machine leaning model and learning the evaluation methods.

E. Practical Outcome (PRo) Implement a Python program using Pandas and Scikit-learn to read a CSV file, load it as a dataset, and print keys, shape, number of rows/columns, first 5 rows, feature names, and description of the data.

F. Expected Affective domain Outcome (ADos):

1. Demonstrate accuracy and attention to detail while inspecting dataset attributes.
2. Show perseverance and problem-solving skills when interpreting dataset metadata.

G. Prerequisite Theory: Scikit-learn provides utilities to load datasets. For custom CSV files, we use Pandas to read the data and can mimic Scikit-learn's Bunch structure using sklearn.utils.Bunch. Key attributes in Scikit-learn datasets:

- data: Feature matrix (numpy array)
- target: Target variable (if supervised)
- feature_names: List of column names (features)
- DESCR: Text description of the dataset

- keys(): Dictionary-like keys of the dataset Common inspection methods: shape, head(), columns, describe(). This step is the first in any ML workflow to understand data distribution and quality.

H. Resources/Equipment Required

Sr. No.	Instrument/Equipment/Components/Trainer kit	Specification	Quantity
1	Computer system with operating system		1
2	Python IDLE or any Python environment with Pandas and scikit-learn installed		1

I. Safety and necessary Precautions followed: a. Turn off power switch only after computer is shut down. b. Do not plug out any computer cables.

J. Procedure to be followed/Source code: Write a Python program to read a CSV file (e.g., iris.csv or any dataset) and using Scikit-learn utilities to print: keys, number of rows-columns, shape, top 5 rows, feature names, and description.

Write code in separate pages

K. Practical related Quiz.

1. Which Scikit-learn utility is commonly used to bundle dataset attributes like data, target, and feature_names?
a) Dataset b) Bunch c) Dict d) DataFrame

2. What does dataset.data.shape return?
a) (n_samples, n_features)
b) (n_features, n_samples)
c) Total number of elements
d) Only number of rows

3. Which Pandas function displays the first n rows of a DataFrame?
a) df.top() b) df.first() c) df.head() d) df.show()

4. In Scikit-learn datasets, what is stored in the 'DESCR' key?
a) Feature values b) Text description of the dataset c) Target values d) Column names

L. References / Suggestions

a. https://scikit-learn.org/stable/datasets.html
b. https://pandas.pydata.org/docs/reference/api/pandas.read_csv.html
c. https://scikit-learn.org/stable/modules/generated/sklearn.utils.Bunch.html
d. https://www.kaggle.com/datasets (for sample CSV files like iris.csv)

M. Assessment:

Obtained Marks
Out of 20

Sign of faculty :

Date: _____

Practical No. 10: Write a Python program that uses Scikit-learn to split a dataset into training and testing sets using train_test_split.

A. Objective: To demonstrate the use of Scikit-learn's train_test_split function to divide a dataset into training and testing subsets, which is a fundamental step in evaluating machine learning models.

B. Expected Program Outcomes (POs): PO1, PO2, PO3, PO4

C. Expected Skills to be developed based on competency: This practical is expected to develop the following skills for the industry-identified competency:

1. Apply Scikit-learn utilities to properly partition datasets for model training and evaluation.
2. Understand the importance of random splitting, stratification, and reproducibility in machine learning workflows to ensure unbiased performance estimation.

D. Expected Course Outcomes (COs) CO2: Practice the NumPy, Pandas, Matplotlib, and scikit-learn libraries' inbuilt functions required to solve machine learning problems. (Partial mapping: Focus on Scikit-learn's data splitting functionality essential for model development and validation.)

E. Practical Outcome (PRo) Implement a Python program using Scikit-learn's train_test_split to split a dataset (e.g., Iris) into training and testing sets with appropriate parameters such as test size, random state, and stratification.

F. Expected Affective domain Outcome (ADos):

1. Demonstrate accuracy and attention to detail while setting parameters for reproducible data splitting.
2. Show perseverance and problem-solving skills when ensuring balanced class distribution in classification tasks.

G. Prerequisite Theory: In machine learning, it is critical to evaluate models on unseen data to estimate real-world performance.

- train_test_split from sklearn.model_selection splits arrays or matrices into random train and test subsets.
- Key parameters:
    - test_size: Proportion of the dataset to include in the test split (e.g., 0.2 for 20%).

- random_state: Controls the shuffling for reproducibility (set to an integer).
- stratify: Ensures the same class distribution in both train and test sets (useful for imbalanced classification). This prevents data leakage and provides reliable performance metrics.

H. Resources/Equipment Required

Sr. No.	Instrument/Equipment/Components/Trainer kit	Specification	Quantity
1	Computer system with operating system		1
2	Python IDLE or any Python environment with scikit-learn and Pandas installed		1

I. Safety and necessary Precautions followed: a. Turn off power switch only after computer is shut down. b. Do not plug out any computer cables.

J. Procedure to be followed/Source code: Write a Python program that uses Scikit-learn to load a dataset (Iris) and split it into training and testing sets using train_test_split.

Write code in separate pages

K. Practical related Quiz.

1. Which function from scikit-learn is used to split data into train and test sets?
a) split_data() b) train_test_split() c) divide_dataset() d) separate()

2. What is the purpose of the random_state parameter in train_test_split?
a) To set the size of the test set
b) To ensure reproducible splits
c) To shuffle only the features
d) To stratify the classes

3. When should you use the stratify parameter?
a) For regression tasks only
b) For classification tasks with imbalanced classes
c) When you want random splitting
d) To increase test size

4. If test_size=0.3 and dataset has 100 samples, how many samples will be in the test set?
a) 30 b) 70 c) 33 d) 25

L. References / Suggestions

a.https://scikitlearn.org/stable/modules/generated/sklearn.model_selection.train_test_split.html
b. https://scikit-learn.org/stable/datasets/toy_dataset.html#iris-dataset
c. https://www.geeksforgeeks.org/how-to-split-a-dataset-into-train-and-test-sets-using-python/

M. Assessment:

Obtained Marks
Out of 20

Sign of faculty :

Date: _____

Practical No. 11: Write a Python program to implement k-Nearest Neighbour supervised machine learning algorithm for given dataset.

A. Objective: To demonstrate the implementation of the k-Nearest Neighbors (k-NN) supervised machine learning algorithm using Scikit-learn on a standard dataset, including data splitting, model training, prediction, and performance evaluation.

B. Expected Program Outcomes (POs): PO1, PO2, PO3, PO4

C. Expected Skills to be developed based on competency: This practical is expected to develop the following skills for the industry-identified competency:

1. Apply Scikit-learn to implement and train a k-Nearest Neighbors classifier/regressor on real-world datasets.
2. Evaluate model performance using appropriate metrics and understand the impact of hyperparameter (k value) selection in supervised learning tasks.

D. Expected Course Outcomes (COs) CO4: Apply supervised learning algorithms based on dataset characteristics.

E. Practical Outcome (PRo) Implement the k-Nearest Neighbors algorithm using Scikit-learn to train a classifier on the Iris dataset, make predictions on test data, and evaluate accuracy.

F. Expected Affective domain Outcome (ADos):

1. Demonstrate accuracy and attention to detail while selecting appropriate k values and interpreting classification results.
2. Show perseverance and problem-solving skills when analyzing model performance and choosing optimal hyperparameters.

G. Prerequisite Theory: k-Nearest Neighbors (k-NN) is a non-parametric, lazy learning algorithm used for classification and regression.

- It predicts the class of a sample by finding the k closest training examples (using distance metrics like Euclidean) and performing majority voting (for classification).
- Key hyperparameters: n_neighbors (k), metric (distance measure), weights (uniform or distance-based).
- Advantages: Simple, no training phase, adapts to any decision boundary.

- Disadvantages: Computationally expensive at prediction time, sensitive to irrelevant features and scale. Feature scaling is often required. k-NN works well when data has locally similar patterns.

H. Resources/Equipment Required

Sr. No.	Instrument/Equipment/Components/Trainer kit	Specification	Quantity
1	Computer system with operating system		1
2	Python IDLE or any Python environment with scikit-learn, Pandas, and Matplotlib installed		1

I. Safety and necessary Precautions followed:

a. Turn off power switch only after computer is shut down.

b. Do not plug out any computer cables.

J. Procedure to be followed/Source code: Write a Python program to implement the k-Nearest Neighbors algorithm on the Iris dataset using Scikit-learn.

Write/attach code in separate pages.

K. Practical related Quiz.

1. Which Scikit-learn class is used to implement the k-NN classifier?
a) KNeighborsRegressor b) KNeighborsClassifier c) NearestNeighbors d) KNNClassifier

2. What is the default distance metric used in KNeighborsClassifier?
a) Manhattan b) Euclidean c) Cosine d) Chebyshev

3. Why is feature scaling important for k-NN?
a) To speed up training
b) Because it is distance-based and features with larger ranges dominate
c) To reduce overfitting
d) It is not needed

4. In k-NN, what does a higher value of k generally lead to?
a) More complex decision boundary
b) Smoother decision boundary and lower variance
c) Faster prediction
d) Higher overfitting

L. References / Suggestions a. https://scikit-learn.org/stable/modules/generated/sklearn.neighbors.KNeighborsClassifier.html b. https://scikit-learn.org/stable/modules/neighbors.html c. https://www.geeksforgeeks.org/k-nearest-neighbors-knn-algorithm/

M. Assessment:

Obtained Marks
Out of 20

Sign of faculty :

Date: _____

Practical No. 12: Write a Python program to implement simple linear regression using Scikit-learn.

A. Objective: To demonstrate the implementation of Simple Linear Regression using Scikit-learn on a dataset, including data preparation, model training, prediction, and evaluation.

B. Expected Program Outcomes (POs): PO1, PO2, PO3, PO4

C. Expected Skills to be developed based on competency: This practical is expected to develop the following skills for the industry-identified competency:

1. Apply Scikit-learn to implement and train a simple linear regression model for predicting continuous target variables.
2. Evaluate regression model performance using appropriate metrics and visualize the regression line, essential for understanding relationships in data and predictive modeling.

D. Expected Course Outcomes (COs) CO4: Apply supervised learning algorithms based on dataset characteristics.

E. Practical Outcome (PRo): Implement Simple Linear Regression using Scikit-learn on a synthetic or real dataset to train the model, make predictions, compute performance metrics, and plot the regression line.

F. Expected Affective domain Outcome (ADos):

1. Demonstrate accuracy and attention to detail while interpreting coefficients, metrics, and visualization of the regression fit.
2. Show perseverance and problem-solving skills when assessing model assumptions and performance.

G. Prerequisite Theory: Simple Linear Regression models the relationship between a single independent variable (feature) $x$and a dependent variable (target) $y$using the equation: $y = \beta_0 + \beta_1 x + \epsilon$where $\beta_0$is the intercept, $\beta_1$is the slope (coefficient), and $\epsilon$is the error term.

- Scikit-learn's LinearRegression fits the model by minimizing the sum of squared residuals (Ordinary Least Squares).
- Key methods: fit(), predict(), attributes: coef_, intercept_.
- Evaluation metrics: Mean Squared Error (MSE), Mean Absolute Error (MAE), R^2 score (coefficient of determination). It assumes linearity, independence, homoscedasticity, and normality of residuals.

H. Resources/Equipment Required

Sr. No.	Instrument/Equipment/Components/Trainer kit	Specification	Quantity
1	Computer system with operating system		1
2	Python IDLE or any Python environment with scikit-learn, Pandas, NumPy, and Matplotlib installed		1

I. Safety and necessary Precautions followed:

a. Turn off power switch only after computer is shut down.

b. Do not plug out any computer cables.

J. Procedure to be followed/Source code: Write a Python program to implement Simple Linear Regression using Scikit-learn on a synthetic dataset.

Write/print Code in separate pages.

K. Practical related Quiz.

1. Which Scikit-learn class is used for Simple Linear Regression?
   a) LogisticRegression b) LinearRegression c) RidgeRegression d) SGDRegressor
2. What does the R^2 score represent?
   a) Proportion of variance explained by the model
   b) Mean squared error
   c) Average absolute error
   d) Model intercept
3. In LinearRegression(), what does model.coef_ return?
   a) Intercept value b) Slope coefficient(s) c) Predicted values d) Residuals
4. A good linear regression model typically has R^2 close to:
   a) 0 b) 1 c) -1 d) Any value between -1 and 1 is acceptable

L. References / Suggestions

a. https://scikit-learn.org/stable/modules/linear_model.html#ordinary-least-squares

b. https://www.geeksforgeeks.org/linear-regression-in-python/

M. Assessment:

Obtained Marks
Out of 20

Sign of faculty :